

PROJECT REPORT

“Hotel360 – Hotel Management CRM System”

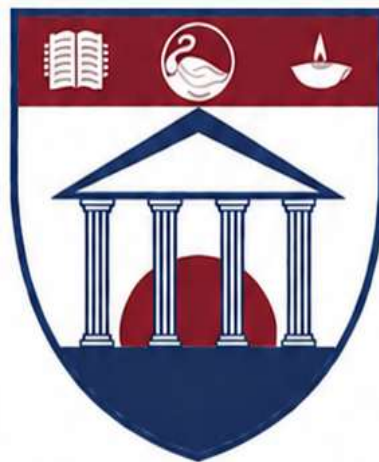
Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirements of the degree of

B.Tech CSE



Submitted By

Name: TEJAS AWASTHI

Section: 4CSE11

Roll No.: CS-23411643

Academic Year 2026–27

CANDIDATE’S DECLARATION

I hereby declare that the project report entitled “Hotel360 – Hotel Management CRM System” is prepared as part of the academic requirements for the Bachelor of Technology in Computer Science and Engineering. The work presented in this report is based on the project material supplied for Hotel360 and has been organized according to the prescribed report format.

I further declare that the information presented in this report is true to the best of my knowledge and that the report has been prepared for academic evaluation. Wherever external technical resources have been used, they have been acknowledged in the References section.

Name: Tejas Awasthi

Roll No.: CS-23411643

Section.: 4CSE-11

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project supervisor for the guidance, support, and constructive feedback provided throughout the development of this project.

I am also thankful to the faculty members and the Department of Computer Science and Engineering for providing the academic environment and resources required to complete this project.

I extend my thanks to my friends, classmates, and family for their encouragement and support during the project work.

Finally, I acknowledge everyone who contributed directly or indirectly to the completion of Hotel360 and its documentation.

INTERNSHIP / PROJECT COMPLETION CERTIFICATE

Certificate id
BGI310091



**BharatGen
Technologies**

CERTIFICATE OF COMPLETION

This is to certify that:

TEJAS AWASTHI

Has successfully completed a 30-day Summer Internship in Web Development.
Worked on the Hotel Management System project during the internship.
Gained practical experience in web development and software implementation.
The internship was completed from **1 June 2026 to 30 June 2026**.
We appreciate the dedication and contribution demonstrated throughout the internship.

PANKAJ KESHARWANI
Co-Founder BharatGen



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Date of issue
5/July/2026

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ABSTRACT

Hotel360 is a web-based Hotel Management Customer Relationship Management system designed to centralize common hotel operations in one application. The system covers room inventory, guest profiles, bookings, billing, room availability, and administrative monitoring.

The project addresses practical problems such as double bookings, fragmented records, manual billing errors, and limited visibility of room occupancy. Server-side date-overlap validation helps prevent conflicting reservations, while automated billing calculates room charges, service charges, taxes, paid amounts, and balances.

The application follows a three-tier architecture using Next.js and React at the presentation layer, Next.js API Routes with Node.js for application processing, and MongoDB with Mongoose for data persistence. Tailwind CSS is used for responsive styling, JWT with HTTP-only cookies provides authentication, and PDFKit supports PDF invoice generation.

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4. PROJECT DESCRIPTION

4.1 Introduction

Hotel360 is a web-based Hotel Management CRM System intended for small- to medium-scale hotel properties. The project brings room management, guest records, booking operations, billing, availability monitoring, and administrative information into a single application.

The project specification identifies fragmented records, double-booking risks, manual billing errors, and limited visibility of room occupancy as key operational challenges.

The system provides operational access for administrators and staff. Core workflows include room configuration, customer management, booking creation and modification, payment recording, invoice generation, and dashboard monitoring.

4.2 Organization Profile

The supplied Hotel360 SRS does not provide a detailed profile of an internship industry or external organization. To avoid introducing unsupported information, the organization profile is provided as an editable insertion section.

Organization Name	Bharatgen Technologies
Project / Internship Role	Hotel360 – Hotel Management CRM System

4.3 Problem Statement

Hotels that depend on paper records, spreadsheets, or disconnected tools can face operational inefficiencies and data inconsistencies.

- Double booking due to absence of centralized availability validation.
- Fragmented guest, room, booking, and billing records.
- Manual billing calculations that can introduce errors.
- Limited visibility of room occupancy and availability.
- Need for controlled access to sensitive operational information.
- Limited consolidated information for management decisions.

Hotel360 addresses these problems through centralized data, server-side booking conflict validation, automated billing, role-based access control, and dashboard-based visibility.

4.4 Project Objectives

- Provide a unified platform for room, guest, booking, billing, and occupancy management.
- Prevent double bookings through server-side date-overlap validation.
- Automate room, service, tax, payment, and balance calculations.
- Provide a date-wise room availability dashboard.
- Maintain guest profiles and booking history.
- Enforce role-based access for administrator and staff users.
- Provide management dashboards for occupancy, revenue, collection, and booking information.
- Generate PDF bills on demand.
- Provide a maintainable foundation for future enhancements.

4.5 Scope of the Project

Hotel360 is scoped as a single-property, browser-based hotel management application.

In-Scope Functions

- Room inventory management.
- Guest/customer management.
- Booking creation, modification, cancellation, and conflict validation.
- Billing, payment-status tracking, and PDF invoice generation.
- Date-wise availability dashboard.
- Administrative dashboard.
- JWT-based authentication and role-based access.

Current Out-of-Scope Functions

- Multi-property operation.
- Live payment gateway integration.
- Automatic email/WhatsApp invoice delivery.
- Drag-and-drop calendar booking.
- Advanced audit timelines and third-party PMS integrations.

4.6 Technologies and Tools Used

Layer / Tool	Technology	Purpose
Frontend	Next.js + React	User interface and navigation
Styling	Tailwind CSS	Responsive styling
Backend	Next.js API Routes + Node.js	API and business logic
Database	MongoDB + Mongoose	Data storage and validation
Authentication	JWT + HTTP-only Cookies	Sessions and roles
PDF Generation	PDFKit	Invoice generation
Package Management	npm	Dependencies and scripts
Version Control	Git	Source-code management

Table 4.1: Technologies and Tools Used

Module	Primary Function
Room Management	Create, edit, view and track hotel rooms.
Customer Management	Maintain guest profiles and booking history.
Booking Management	Manage reservations and prevent date conflicts.
Billing	Calculate charges, payments, balances and PDF invoices.
Availability Dashboard	Display room status by date.
Administrative Dashboard	Show occupancy, revenue and booking summaries.

Table 4.2: Major Hotel360 Functional Modules

4.7 System Architecture

Hotel360 follows a three-tier architecture consisting of the presentation tier, application/business tier, and data persistence tier.

PRESENTATION TIER	APPLICATION / BUSINESS TIER	DATA PERSISTENCE TIER
Next.js + React Tailwind CSS	Next.js API Routes Node.js Authentication and business rules	MongoDB Mongoose ODM
UI and navigation	REST APIs, validation, authorization, booking and billing	Rooms, customers, bookings, billing, users

Figure 4.1: Three-Tier System Architecture

The data model contains Room, Customer, Booking, Billing, and StaffUser entities. Customer-to-Booking and Room-to-Booking are one-to-many relationships, while Booking-to-Billing is one-to-one.

ROOM	CUSTOMER	BOOKING
roomId (PK) roomNumber type price/status	customerId (PK) name/email/phone ID proof details	bookingId (PK) customerId (FK) roomId (FK) dates/services/status
Relationship		Cardinality
Customer → Booking		1:M
Room → Booking		1:M
Booking → Billing		1:1

Figure 4.2: Entity-Relationship Model

External Entity	Interaction with Hotel360
Administrator / Staff	Login, room data, guest data, bookings and payments; receives dashboards, confirmations and invoices.
Guest / Customer	Personal/contact information recorded by staff; receives booking and billing information.
MongoDB Database	Stores rooms, customers, bookings and billing; returns records and query results.

Figure 4.3: Context-Level Data Flow

4.8 Methodology

The project follows a structured, iterative software engineering approach combining requirements analysis with agile development principles.

Requirements	Design	Implementation	Testing	Deployment
Identify problems	Architecture	Frontend	Unit	Build
Define users	Data model	Backend	Integration	Server
Requirements	API design	Database	Security	Database
			UAT	HTTPS

Figure 4.4: Project Development Methodology

4.8.1 Requirement Analysis

Identify hotel operational problems, user roles, and functional and non-functional requirements.

4.8.2 System Design

Define architecture, database models, APIs, authentication, and booking conflict validation.

4.8.3 Implementation

Develop the frontend, backend, database operations, authentication, and invoice generation.

4.8.4 Testing and Integration

Perform unit, API, end-to-end, security, browser compatibility, usability, and acceptance testing.

4.8.5 Deployment

Prepare the production build, server, MongoDB hosting, HTTPS, and secure environment configuration.

4.9 Expected Outcomes

- Centralized hotel operational data.
- Reduced double-booking risk.
- Improved billing accuracy.
- Faster room-status assessment.
- Controlled access through role-based authentication.
- Improved management visibility through dashboards.
- Professional PDF invoices.
- A modular foundation for future enhancements.

The current version is intended for small- to medium-scale, single-property hotel operations.

4.10 Certificates of Completion and Communication Proof.

Attachment A – Industry / Organisation Completion Certificate



**BharatGen
Technologies**

CERTIFICATE OF COMPLETION

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Attachment B – Communication / Mail Proof

Attach relevant official communication, completion mail, or approval proof here



BHARATGEN TECHNOLOGIES
Empowering the New Generation of Bharat



OFFER LETTER

Ref No: BGIMA20260000154

May 26, 2026

Dear **Tejas Awasthi**,

We are pleased to inform you that you have been selected for an internship at **BharatGen Technologies** as a **Web Development Intern**.

The internship will be for a period of one month, from **1 June 2026 to 30 June 2026**. During this period, you will be working on the Hotel Management System project and gaining practical experience in web development and software implementation.

We look forward to your valuable contribution and wish you a successful learning experience with us.

Thank you, and congratulations!

Regards,
BharatGen Technologies



Head Office: Soraon, Prayagraj, 212502

Ref No: BGIMA20260000154

May 26, 2026

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